# Bits, Hyperbits and Landauer

## Information Physics within the FFGFT Framework

*Fundamental Fractal Geometric Field Theory (T0 Time-Mass Duality)*

*Landauer, Hyperbit and Scale Anchor*

Johann Pascher

ORCID: 0009-0000-6518-4064

https://github.com/jpascher/T0-Time-Mass-Duality

September 2026

Version 1.0 — Kindle Edition, English Version

*For all those who do not take the bit
for granted.*

# Contents

## Preface

> The region comes first. The description comes afterwards.
>
> A271, Landauer and Phase-Space Bookkeeping

This book emerged from an encounter: Douglas Matzke, in his contribution ‘Black Holes from Hyperbits’, reveals the entire physics of black holes from a single Clifford algebra axiom — $e_i^2 = +1$. No general relativity, no free parameters. The mass of a bit follows from Landauer’s principle and $E = mc^2$ as a universal number.

The FFGFT (Fundamental Fractal Geometric Field Theory) arrives at the exactly opposite insight: There is no *universal* bit mass. The minimal information unit is the winding quantum $\Delta w = 1$ on the fundamental torus $T^4$, and its energy is $E_{\text{bit}} = \hbar c/L$ — dependent on the physical scale $L$. Landauer’s $k_B T \ln 2$ is the special case of a specific scale and temperature, not a universal law.

This tension is productive. It forces a clean separation of three things that regularly merge in the literature:

1. The **physical region** in phase space — measurable in units of $h^f$, existing independently of any naming.
2. The **description** of this region — bit, qubit, analog value: decisions of the observer, not constants of nature.
3. The **thermodynamics of erasure** — Liouville theorem, not semantics: physics counts regions and does not read them.

The book collects and organizes the relevant FFGFT primary documents. It is not a popularizing simplification — the derivations are included. The status of each statement is explained in the appendix. The primary documents as PDFs are freely accessible at `https://github.com/jpascher/T0-Time-Mass-Duality`.

*Linz, September 2026* Johann Pascher

# Part I

# The Foundation: What Is a Bit?

# Chapter 1

## The Bit as a Physical Object

A bit is not a thing. It is a decision.

### 1.1 What is a Bit Really?

The intuitive answer: a bit is a $0$ or a $1$. That is a descriptive decision, not a physical statement.

The physical statement is different: A bit is a physical system with two distinguishable macro-regions in phase space. [B] Whether one calls these regions $\{0, 1\}$, $\{\uparrow, \downarrow\}$ or $\{$left, right$\}$ changes nothing about the physics. The naming is a descriptive decision of the observer — not a law of nature.

***Intuition:*** *Imagine two glasses on a table. One is full, one is empty. These are two distinguishable physical states — a bit, before you call it that. Whether you say "full = 1, empty = 0" or vice versa, does not change the physics. The naming is your decision.*

**Core Distinction**

**Physical Region $\neq$ Description of the Region** [SETZUNG]
The region in phase space exists independently of how one divides it. The naming as "bit" is an external interpretation.

### 1.2 Multiple Realization: The Same Bit, Different Carriers

A bit with the value $1$ can be realized by fundamentally different physical systems: a CMOS transistor at 5 V, a magnetic spin $\uparrow$, a circularly polarized photon, an abacus bead on the left of the working field, a nanomagnet pillar with orientation $+z$.

The information content is identical in all cases: 1 bit. The energies vary by many orders of magnitude. It follows: Energy is a property of the **carrier**, not of the information. [B]

***Intuition:** A letter contains the same information whether you send it by airmail or by e-mail. The energy of transmission is completely different — the information content is identical. Energy clings to the carrier, not to the information.*

## 1.3 The Three Levels of Description

In the FFGFT, three levels are strictly separated (Doc. A272): [B]

**Level 1 — Construction.** The engineer determines during chip design which region operations physically exist — bijective or non-bijective. This is cast in silicon, unchangeable during operation. This decision determines the thermodynamic cost structure — not the program.

**Level 2 — State Transition.** During operation, state transitions take place in a clocked manner. Landauer costs arise *only* for non-bijective region mappings. The clock frequency, not the computational operation, determines the entropy flow.

**Level 3 — Description.** The program chooses which hardware is used. That is resource management. The thermodynamic costs still arise — determined by construction and switching operation, independent of what the description level does with the results.

## 1.4 How Many States Are in a Region?

A colloidal particle in a 1-µm trap contains physically about $7{,}9 \cdot 10^9$ orthogonal quantum states. If one names roughly two zones "left" and "right," one describes this region with 1 logical bit. If one divides it into 1000 zones, one describes the same region with $\sim 10$ logical bits. The physical region is identical in both cases — the depth of description is a choice of the observer. [B]

The counting basis is always $h^f$: the fundamental discretization of the $f$-dimensional phase space, independent of naming and depth of description.

## 1.5 Bit, p-Bit, Qubit: A Scale

The classical bit operates in a deep potential well $U \gg k_B T$ — macroscopically stable, for years. The qubit requires metastable coherence in the millisecond range at mK temperatures. In between lies the **p-bit**: a magnetic nanostructure in which the torsion barrier becomes thermally surmountable.

The FFGFT shows that the natural scale for p-bits is geometrically predetermined: Level $N = 8$ of the $\xi$ hierarchy yields $L_8 \approx 21{,}6$ nm — exactly the transistor limit size. [K Doc. 184]

***Intuition:*** *Three operating points, one scale. The classical bit is like a heavy ball deep in a valley — stable for years. The qubit is a ball on a mountain peak, held there by deep cooling. The p-bit lies in between: the valley is flat enough that heat occasionally kicks the ball to the other side. The FFGFT geometry predicts where this boundary lies: at 22 nanometers.*

These three regimes — classical bit, p-bit, qubit — are not separate worlds. They are three operating points on the same physical scale. [B]

# Chapter 2

## Geometry or Information? — FFGFT in the Information Formalism

> That FFGFT can be mapped into an information formalism does not prove that information is the prior. It shows that both descriptions are projections of the same structure.
>
> Doc. 243

### 2.1 The Fundamental Question

Is information or geometry the more fundamental primitive? This question is controversially discussed in the IPI context — Onur Teker (UIFT) and others advocate information as the ontological foundation from which geometry *follows*.

FFGFT gives a precise answer: Both descriptions are **projections of the same structure**. Which one chooses does not decide the physics — but it does decide what is lost in the projection.

### 2.2 The Geometric State Space in the FFGFT

The fundamental state space in FFGFT is the $T^4$ torus with winding numbers $(n_\theta, n_\varphi) \in \mathbb{Z}^2$ and phase $\theta_4 \in [0, 2\pi)$. A physical state $\Psi$ is completely characterized by:

$$\Psi = \left(n_\theta,\ n_\varphi,\ \theta_4,\ \varepsilon\right)$$

where $\varepsilon \approx 0{,}001$ is the non-closure (Doc. 193).

### 2.3 The Information Projection $\pi_I$

The projection $\pi_I$ maps the geometric state onto an information representation:

$$\pi_I : \ \Psi(n_\theta, n_\varphi, \theta_4, \varepsilon) \ \longmapsto \ \left(I(n_\theta, n_\varphi),\ \mathrm{sgn}(\theta_4),\ \langle\varepsilon\rangle_{\text{therm}}\right)$$

where $I = \log_2 |\mathscr{W}|$ is the number of bits of admissible winding configurations.

**Decisive:** $|\mathscr{W}|$ must be finite for $I$ to be well-defined. The $T^4$ geometry provides this bound via the maximum admissible winding number $n_{\text{max}} \sim 1/\xi$. Without this geometric bound, $I$ diverges — the bit counting *presupposes* the geometric state space, it does not create it.

## 2.4 What is Lost in the Projection

The mapping $\pi_I$ is not a bijection. Three pieces of information are lost:

1. **Continuous phase** $\theta_4 \in [0, 2\pi)$ is reduced to binary $\text{sgn}(\theta_4)$ — loss of a real degree of freedom.
2. **Winding numbers** $(n_\theta, n_\varphi)$ are projected onto a scalar bit number $I$ — loss of directional information.
3. **Non-closure** $\varepsilon$ is thermally averaged — loss of deterministic fine structure.

**Kernresultat**

Information *as such* is not an ontological primitive in FFGFT — it is a useful description that emerges from the geometric state space during projection. Wheeler's "It from Bit" reverses the derivation: FFGFT says "Bit from It" — where "It" is the $T^4$ torus.

## 2.5 Consequence for the Landauer Discussion

That $\pi_I$ is not a bijection has a direct consequence for Landauer: The erasure of a bit corresponds to a region reduction in the geometric state space, not in an abstract information space. Thermodynamics counts winding states — it does not know bits. Bits are a descriptive decision about this state space.

*Open Question* (Doc. 243): Is there a complete reconstruction $\pi_I^{-1}$ that uniquely infers the geometric state from the information representation? If not — and this is the FFGFT thesis — then geometry is the deeper primitive.

# Chapter 3

## Landauer and the Phase Space Accounting

> Two glasses stand on a table. Next to them, the world ocean. One pours one of the glasses into the ocean. The glass volume does not disappear — it distributes over 1.4 billion cubic kilometers.
>
> A271

### 3.1 The Image That Carries Everything

Landauer's result can be formulated without the concept of entropy: A physical system occupies a **region in phase space**. A mapping that reduces several initial regions to one target region shrinks this volume. The **Liouville theorem** forbids volume shrinkage in a closed system. Therefore a bath must take over the volume, and the price is heat. [B]

**Landauer Limit (Physical Formulation)**

$$Q \geq k_B T \ln\left(\frac{W_{\text{before}}}{W_{\text{after}}}\right)$$

$W_{\text{before}}$ and $W_{\text{after}}$ are physical region sizes in units of $h^f$. The ratio is $\ln 2$ only as a binary special case. Counting is by distinguishability, not by semantics. [B]

***Intuition:*** *Imagine a needle in a huge haystack. If you reduce the haystack to half, the other half must go somewhere. The bath — the environment — takes it up, and the energy for this is heat. That is Landauer's principle, without information, without semantics: only volume that must go somewhere.*

## 3.2 What Landauer Actually Wrote in 1961

Landauer explicitly wrote 'logically irreversible' — logically irreversible, not physically irreversible. He meant: a mapping in which different *logical* input states lead to the same logical output state. That is a statement about descriptions, not about physical regions. [Q]

The step from the carrier operation to the universal statement about 'information as such' is not found in Landauer's original work — it was accomplished through reception (Doc. A272).

## 3.3 Liouville is the Reason — Not Ignorance

The epistemic reservation must be explicitly booked: **Not measurable does not mean not present.** The Landauer limit applies because of mechanics (Liouville theorem), not because of the observer's ignorance. The irreversibility is statistical, not ontologically absolute. [B]

This means: Reversible computing (Bennett, Fredkin) is in principle conceivable without Landauer costs. The costs arise from the hardware decision for non-bijective region mappings — not from computing itself.

***Intuition:*** *Pouring a glass of water into the ocean is irreversible — not because physics forbids it, but because it is practically impossible to find all water molecules again and pour them back. Landauer costs arise in the same way: not from a law of nature that forbids reversibility, but from the decision to merge regions instead of rearranging them bijectively.*

## 3.4 Content Neutrality

Thermodynamics **counts regions and does not read them**. [B] Whether the erased bit meant "true" or "false," whether it belonged to a correct computational step, whether it was read at all — thermodynamics does not know this level. It knows only: region A and region B exist; after the region reduction, target region Z exists; the bath takes up the difference.

This is called **content neutrality of thermodynamics**.

## 3.5 Shannon and Boltzmann: Two Separate Bookkeepings

Shannon describes how much uncertainty a *description* carries:

$$H = -\sum_i p_i \log_2 p_i$$

a quantity about descriptions. Boltzmann describes how large a physical *region* is:

$$S = k_B \ln W$$

a quantity about regions. The two are identical only when the description is mapped exactly onto the physical microstates and these are uniformly distributed. In every other case they are different things with different units. [B]

***Intuition:*** *Shannon asks: How much do you not know? Boltzmann asks: How large is the region? Both questions have numbers as answers, and both numbers can coincidentally be equal — but they are not the same. It is like the question of whether the temperature in Vienna coincidentally has the same number as the population of a small town: numerical agreement, completely different meaning.*

## 3.6 DRAM Refresh and Qubit Decoherence: Continuous Entropy Flow

A real computer does not erase discrete bits — it carries a **continuous entropy flow**. DRAM refresh, qubit decoherence: Both are non-bijective region mappings that occur clocked, independent of what is computed. Clock gating and power gating are hardware decisions at Level 1, not program decisions. [B]

**To Remember**

Landauer costs run with the clock frequency — practically constant as long as the computer runs. CPU utilization says little about this: It measures the description level, not the region level.

## 3.7 Connection to the FFGFT

The Landauer bookkeeping lives at the level of statistical mechanics — it has no direct contact with the $\xi$ scheme of the FFGFT. This is an explicit statement of Doc. A271: the own bookkeeping of this level is independent.

What FFGFT adds: the insight that $k_B T \ln 2$ is not a universal law, but the special case of a specific scale and temperature. At the Planck scale, the Landauer limit is of the order $E_P$; at macroscopic scales, it is the familiar thermodynamic quantity. The scale makes the difference (Ch. 9). [K]

# Chapter 4

## State vs. Process — Information Is Not Energy

> The mere storage of information is energy-neutral. Only processing has an energy dimension.
>
> Doc. 247

### 4.1 The Question

In information physics — represented by Wheeler ("It from Bit"), M. M. Vopson **[V1–V3]** and others — information is often treated as an ontological primitive, from which energy and matter follow. Landauer's principle is cited as evidence: Since erasing a bit costs energy, information is physically real.

This conclusion confuses two structurally different concepts.

**Kernresultat**

**Information as state** — a configuration, a stored bit, a winding number — exists without energy expenditure. It is a topological or geometric property.

**Information as process** — reading, writing, erasing, transforming — costs energy. Landauer's principle applies *only* to erasure, thus to a process, not to the state itself.

### 4.2 Empirical Evidence: The Permanent Memory

Anyone who flattens the difference must accept a direct empirical consequence: Every stored bit on a switched-off device would continuously consume energy — which would make permanent memory fundamentally impossible. The existence of flash memory, magnetic hard drives and ROM chips refutes the equation of information and energy.

A permanent memory holds millions of bits over years at nearly zero energy consumption. It is a counter-experiment to any theory that equates information and energy.

## 4.3 The FFGFT Reading

In FFGFT, states are geometric configurations on $T^4$: winding numbers $(n_\theta, n_\varphi) \in \mathbb{Z}^2$. Energy is a derived quantity that follows from *processes* — from the transition between configurations — not from the mere existence of a configuration.

The rest mass of a particle in FFGFT is:

$$m = \frac{\hbar c}{L} \cdot n \qquad (n \in \mathbb{Z},\ L \text{ scale})$$

The winding number $n$ is a state — it costs no energy as long as the topology remains intact. Only the transition $n \to n'$ — a process — requires or generates energy.

## 4.4 M. M. Vopson and the FFGFT Restriction

M. M. Vopson postulates in **[V1]** a mass-information equivalence: Every bit carries a universal rest mass. That is attractive — and according to FFGFT false.

FFGFT restriction: Vopson's bit mass would be system-invariant. FFGFT shows that the energy of a bit $E_{\text{bit}} = \hbar c/L$ depends on the scale $L$ (Doc. 257). Bits on different scales have different energies. There is no universal bit mass — neither according to Landauer (Doc. A271–A272) nor according to FFGFT.

**FFGFT**

**Wheeler:** "It from Bit" — geometry from information.
**M. M. Vopson [V1,V2]:** Bit mass universal — information has rest mass.
**FFGFT:** "Bit from It" — information as projection from geometry; bit mass scale-sensitive; storing is free, erasing costs.

## 4.5 What Landauer Really Shows

Landauer's principle is, in this perspective, not evidence for the physical reality of information as an ontological primitive. It is evidence for the physical reality of the *carrier*: When a carrier is irreversibly reconfigured, costs arise. The information on the carrier plays no role in this — thermodynamics counts regions, not meanings (Doc. A271).

# Chapter 5

## Carrier and Information — A Necessary Distinction

> The popular short version exceeds what Landauer's proof shows.
>
> A272

### 5.1 The Question

Since Landauer's work of 1961, it has been claimed: Every erased bit must at least $k_B T \ln 2$ heat. This generalization goes beyond what Landauer's proof shows. Doc. A272 provides the textual criticism; the physical substance is in A271.

### 5.2 Two Definitions

**Information** is an abstract quantity. A bit is a unit of information. Information has no mass, no energy, no entropy and no physical localization.

**Carrier** is a physical system that represents information. The carrier has energy, entropy and other physical properties.

This separation is a stipulation. It is not proven; it determines what is talked about. The entire yield depends on it being maintained.

### 5.3 Dictionary A271/A272

Both documents make the same distinction in different words:

| A271 (Bookkeeping) | A272 (Textual Criticism) | Level |
|---|---|---|
| Region in phase space | Carrier | physical |
| Description | Information | abstract |
| Region reduction $\{A, B\} \to Z$ | RESTORE TO ONE | physical |
| Non-bijectivity | State space reduction | physical |
| Descriptive decision | Interpretive erasure | abstract |

## 5.4 Energy Depends on the Carrier

**Satz 5.1.** *An information bit cannot be assigned any energy, only its carrier.*

*Proof.* Suppose a bit had intrinsic energy. This would have to be independent of the physical realization. A bit $= 1$ can however be realized by CMOS (5 V), by a magnetic spin or a photon — with identical information content, but different energies. Therefore energy is not a property of the information, but of the carrier.

The argument is a multiple-realization argument and carries exactly as far as the premise holds: A bit remains the same bit when the carrier changes. Where this invariance breaks (Ch. 20), the proof must be supplemented.

## 5.5 Hypostatization as a Linguistic Mechanism

When one speaks of ‘the bit’, one treats the abstract category as if it had its own physical existence — that is **hypostatization**. Digital technology structurally favors this error: The word ‘bit’ denotes both the hardware unit (carrier) and the information unit (abstract). A language without this ambiguity would make the error more difficult.

## 5.6 Scope and Limits of the Statement

What is not disputed here: Landauer's calculation is correct for the operation he considers — the physical RESTORE-TO-ONE operation on a carrier in thermal equilibrium. What is disputed is the applicability of this result to abstract information.

# Part II

# The Hyperbit: Geometry from Distinguishability

# Chapter 6

## Matzke's Hyperbit Idea

> From 'there are bits' follow dimension, three colors, one fermion generation, the Schwarzschild radius.
>
> Doc. 326

### 6.1 The Starting Point: A Sign

Matzke begins with $e_i^2 = +1$. That is a **bit** — something that yields one when squared. No space, no time, no metric. Only the statement: there are distinguishable things. [SETZUNG]

***Intuition:*** *Imagine you know only one thing: There are things that differ. Nothing more. No space, no force, no matter — only distinguishability. Matzke's hyperbit is exactly that: the mathematical object that carries precisely this statement, and nothing more.*

**Matzke: Axiom and Bridges**

[SETZUNG] Single axiom: $e_i^2 = +1$ (Hyperbit).

Physical units via three external bridges:

- Bekenstein: $4\ell_P^2$ per bit (entropy-area relation)
- Compton = Schwarzschild: $\ell_P$ as length scale
- Landauer-Einstein: $m_{\text{bit}} = m_P \ln 2/(2\pi)$

### 6.2 Why Cl(6) — and Why Three Colors

If one takes several $e_i$ and lets them anticommute, a structure with dimension, orientation and rotations emerges from nothing. Cl$(6)$ decomposes into exactly three Witt pairs — three packages each consisting of one creator and one annihilator. [B]

According to Furey, this is exactly the structure of a fermion generation: three colors, one lepton. The three is not chosen, it falls out. **Geometry as bookkeeping over distinguishability** — not as a presupposition.

***Intuition:*** *You take three coins and place them so that none touches the other. Already you have a triangular structure — not because you drew a triangle, but because the coins force it. Likewise: The Clifford algebra* Cl$(6)$ *generates threefold structure not because one builds it in, but because anticommutativity plus six generators leaves nothing else.*

## 6.3 The Black Hole Part

Bekenstein: A black hole has entropy proportional to the area, one bit per four Planck areas. Matzke reverses: If a bit is an area, then a collection of bits is a volume. He sets the Compton wavelength equal to the Schwarzschild radius and obtains the Schwarzschild radius *without ever using the Einstein equations*. [B]

For $20\,M_\odot$: $59{,}1$ km. Exactly correct.

The stability threshold is at $6{,}41$ bits. Cl$(6)$ has $6{,}000$ bits. The universe consists of fermions that sit exactly $0{,}41$ bits below the collapse threshold.

***Intuition:*** *Imagine you stack Lego bricks. Above a certain height, the tower collapses under its own weight. In Matzke's framework, a system of bits collapses into a black hole above a certain bit count. The critical number: 6.41. A fermion has 6.00 bits — just barely below.*

## 6.4 Where the Elegance is Bought

The $6{,}41$ depends on a stipulation: To get from bits to masses, Matzke needs a conversion rate. He takes Landauer ($E = k_B T \ln 2$) and evaluates at Planck temperature:

$$m_{\text{bit}} = \frac{m_{\text{P}} \ln 2}{2\pi} \approx 0{,}11\, m_{\text{P}}$$

A *universal* bit mass. [SETZUNG]

Only: Landauer is a statement about thermal carrier ensembles, not about abstract information (A271–A273). And the Planck temperature is not the temperature of any physical system

— it is an energy scale. The choice $T = T_P$ is a stipulation that does not justify itself.

**Shortest Comparison**

Matzke's algebra is beautiful because it shows how far one gets with pure combinatorics. It is *simple* because it stops asking at three points — three borrowed bridges.

FFGFT pays the price upfront ($\xi$ derive, $L_0$, justify the torus) and then needs no bridges. Both calculations work out — they look elegant at different points.

# Chapter 7

## The p-Bit — Between Bit and Qubit

> The thermal noise of the p-bit is not fundamental randomness. It is the macroscopically unresolved sampling of deterministic torsion configurations.
>
> Doc. 184

### 7.1 The Operating Point Between Bit and Qubit

The classical bit requires a deep potential well $U \gg k_B T$ — robust over years, without cooling. The qubit operates at the opposite extreme: metastable torsion, coherence in the millisecond range, enforced by cooling to mK temperatures.

The **p-bit** occupies the conceptually previously unexplored intermediate range: a magnetic nanostructure in which the torsion barrier becomes thermally surmountable. The system switches spontaneously between the two macrostates.

### 7.2 The Natural Scale: $L_8 \approx 22$ nm

The $\xi$ length hierarchy of the FFGFT (Doc. 173) defines geometrically distinguished scales:

$$L_N = L_0 \cdot \left(\frac{1}{\xi_{\text{par}}}\right)^N, \qquad \xi_{\text{par}} = \frac{4}{30000}$$

Level $N = 8$ yields $L_8 \approx 21{,}6$ nm — simultaneously excitonic Bohr radius, transistor limit size and fine structure constant. This is not an approximation, but a geometric derivation. [K Doc. 184]

At this scale, the torsion barrier is so low due to the Stoner-Wohlfarth energy $U = K_u \cdot V$ that thermal transitions occur regularly.

***Intuition:*** *Imagine a ball on a flat hilltop. On a high mountain (classical bit), it takes a strong push to reach the other side. On a low hill (p-bit), a gust of wind is enough — the normal thermal energy of the environment.*

*The only question is: How high exactly must the hill be? FFGFT says: It is naturally flattest at 22 nanometers.*

**Thesis: Doc. 184**

The thermal noise of the p-bit is not fundamental randomness. The transitions follow the sub-Planck-scale geometry on the time scale $t_0 = t_P/7500$. The stochasticity is epistemic — it reflects unresolved geometry, not ontological chance. [S Doc. 184]

## 7.3 Applications and Outlook

p-Bits are an active field of research in spintronics. The FFGFT prediction: p-bit nanostructures should find their optimal operating temperature-to-barrier relation at $\sim 20$ nm. [K]

This is falsifiable: If the optimal material scale systematically deviates from 22 nm, the geometric derivation must be reviewed.

# Chapter 8

## Spin, Qubit and State Spaces in the FFGFT

> Coherence is not magic — it is a question of the barrier height relative to $k_B T$.

---

### 8.1 Spin as Torsion Configuration

In the FFGFT, spin is not an abstract property — it is a discrete torsion configuration in the fundamental torsion field. Spin-up and spin-down are geometrically distinguished states, stable due to the depth of the potential well $U \gg k_B T$. [SETZUNG]

***Intuition:*** *A permanent magnet holds its orientation for years — that is spin in the classical limit. The spin structure is built into the geometry of the electrons, not added as a separate axiom.*

### 8.2 The Qubit: Metastable Torsion

The qubit operates in the superposition state of two torsion configurations. Coherence means: The phase relationship between $|0\rangle$ and $|1\rangle$ is preserved. Decoherence means: The bath takes over the torsion volume of the superposition. [B]

This is a Landauer process: The region reduction from $\{|0\rangle, |1\rangle, \text{superposition}\}$ to a classical state costs volume, which the bath takes up. **Decoherence is Landauer.** [B]

### 8.3 State Spaces in the FFGFT

The Hilbert space of the qubit corresponds to the space of torsion winding modes on $T^4$. Each winding number $w \in \mathbb{Z}$ is a distinguished state. Superposition means: coherent superposition of several winding numbers.

The energy of a state with winding number $w$ on scale $L$:

$$E_w = \frac{w \cdot \hbar c}{L}$$

This is the FFGFT expression for the quantization condition — not an approximation, but an exact geometric relation (Doc. 257).
[K]

# Part III

# Landauer within the FFGFT

# Chapter 9

## Information Unit and Scale

> $k_B T \ln 2$ is not a universal law. It is the special case of a specific scale and temperature.
>
> Doc. 257

### 9.1 The Question

Where does the number $k_B T \ln 2$ in Landauer's formula come from? Why exactly this combination of temperature, Boltzmann constant and the logarithm of 2? And does it apply everywhere — or only under certain conditions?

The FFGFT gives a precise answer: It applies exactly when the geometric energy of a winding quantum on the considered scale coincides with the thermal energy of the bath. That is a condition, not a universal constant.

The complete derivation follows in three levels. [B Doc. 257]

### 9.2 Level 1: The Winding Quantum — Topological, Universal

**Minimal Information Unit — [SETZUNG]**

The minimal information unit in the FFGFT is the **winding quantum** $\Delta w = 1$ on the fundamental torus $T^4$.
The homotopy group of the torus is $\pi_1(T^4) \cong \mathbb{Z}^4$ — it knows only integer winding numbers. A halved winding quantum does not exist. This discreteness is a topological fact, not a postulate. [Q]

*For external support:* $\pi_1(T^n) \cong \mathbb{Z}^n$ is a standard result of algebraic topology. It follows from the theorem that the $n$-dimensional torus $T^n = S^1 \times \cdots \times S^1$ ($n$-fold product of circles) is, and that $\pi_1(S^1) \cong \mathbb{Z}$ (the circle has winding number $\in \mathbb{Z}$), and that the fundamental group of a product space is the product of the fundamental groups:

$$\pi_1(T^4) = \pi_1(S^1)^4 \cong \mathbb{Z}^4.$$

Textbook references: Hatcher, *Algebraic Topology* [**?**], Ch. 1.2 (freely available: `https://pi.math.cornell.edu/~hatcher/AT/AT.pdf`); Bredon, *Topology and Geometry* [**?**], Ch. III.§5. The FFGFT-specific argument — identification of $\Delta w = 1$ with 1 bit — is the conceptual stipulation; the integrality itself is external, established mathematics. [Q]

The identification $\Delta w = 1 \leftrightarrow$ 1 bit is the conceptual bridge between topology and information theory. It is **scale-independent**: Whether on the Planck scale or the cell scale — a winding quantum is a winding quantum. [B]

***Intuition:*** *Imagine a string wrapped once around a post. You can wrap it twice or three times — but not one and a half times without it coming undone. That is $\pi_1(T^4) \cong \mathbb{Z}^4$: winding numbers are integers, enforced by the topology of the torus. A bit corresponds to exactly one such winding.*

## 9.3 Level 2: The Bit Energy — Geometric, Scale-Dependent

A winding quantum on a physical scale $L$ has an energy. This energy follows from two known principles:

**Step 1 — Action of the winding quantum.** The action of a winding quantum is $S = \Delta w \cdot h = h$. That is a direct consequence of the quantization condition on $T^4$.

**Step 2 — Momentum from the uncertainty relation.** On a spatial scale $L$, $\Delta x \cdot \Delta p \sim \hbar$, so $p \sim \hbar/L$.

**Step 3 — Energy of the winding quantum.** With $E = pc$ (massless mode):

$$\boxed{E_{\text{bit}}(L) = \frac{\hbar c}{L}}$$

That is the energy of a minimal information quantum on scale $L$. It depends on the scale — and only on it. [B]

***Intuition:*** *The formula $\hbar c/L$ is familiar from physics: That is the energy of a photon with wavelength $L$. A winding quantum on the torus behaves energetically like such a photon — its energy is determined by the spatial scale on which it exists.*

## 9.4 Level 3: Landauer as a Special Case — Thermodynamic, One Scale

Landauer's formula $E_L = k_B T \ln 2$ applies in the thermodynamic regime. This is exactly the case when the geometric bit energy coincides with the thermal energy of the bath:

$$E_{\text{bit}}(L) = k_B T \ln 2 \quad \Longleftrightarrow \quad \frac{\hbar c}{L} = k_B T \ln 2$$

Solving for the scale yields the **equivalence temperature**:

$$T_{\text{equiv}}(L) = \frac{\hbar c}{k_B \, L \, \ln 2}$$

That is the temperature at which Landauer's formula applies for a given scale $L$ — and *only* at this temperature. [B]

## 9.5 The Consequence: 70 Orders of Magnitude Variation

Inserting various physical scales:

| Scale $L$ | $E_{\text{bit}}(L)$ | $T_{\text{equiv}}$ [K] |
|---|---|---|
| Planck ($10^{-35}$ m) | $E_P \approx 1{,}96$ GJ | $2 \times 10^{32}$ |
| Proton ($10^{-15}$ m) | $\sim 200$ MeV | $3 \times 10^{12}$ |
| Atom ($10^{-10}$ m) | $\sim 2$ keV | $3 \times 10^{7}$ |
| **Cell** ($10^{-5}$ m) | $\sim 0{,}02$ eV | $\approx$ **331** |
| p-Bit (22 nm) | 9,2 meV | $\approx 100$ |
| Cosmic ($10^{26}$ m) | $\sim 10^{-51}$ J | $10^{-30}$ |

Landauer's $k_B T \ln 2$ at room temperature ($T = 300$ K) is $\approx 2{,}9$ meV — that corresponds to a scale of $\approx 48\,\mu$m, near the cell scale. [K]

***Intuition:*** *The table shows: The equivalence temperature varies over* 70 orders of magnitude. *That means: Landauer's $k_B T \ln 2$ applies exactly only at a single scale for a given temperature — not everywhere. It is not a universal law, but a hit in the $E_{bit}(L)$ spectrum.*

## 9.6 The Hierarchy — Summarized

$$T^4\text{-Geometry} \longrightarrow \Delta w = 1 \longrightarrow E_{\text{bit}}(L) = \frac{\hbar c}{L}$$
$$\longrightarrow E_L = k_B T \ln 2 \text{ (special case)}$$

Geometry is more primary than energy. Energy is a scale-dependent projection of the $T^4$ geometry, not its foundation. [B]

Matzke's approach sets $T = T_P$ and obtains:

$$m_{\text{bit}} = \frac{m_P \ln 2}{2\pi} \approx 0{,}11\, m_P$$

That is the hit at the Planck scale — a stipulation that one can also calculate in the FFGFT picture: $E_{\text{bit}}(L_P) = E_P$. The difference: Matzke declares this hit to be universal. FFGFT shows that it is scale-sensitive. [K]

## 9.7 What is Proven and What is Open

**Proven [B]:** The hierarchy topology → geometry → thermodynamics is internally consistent. The derivation $E_{\text{bit}}(L) = \hbar c/L$ from winding quantization and uncertainty relation is mathematically correct. Landauer's formula is a special case in this hierarchy.

**Derived and numerically verified [K]:** The equivalence temperature matches biological room temperature at the cell scale. The p-bit prediction $L_8 \approx 22$ nm is consistent.

**Open [S]:** Is there a direct experimental test that distinguishes $E_{\text{bit}}(L) = \hbar c/L$ from Landauer's $k_B T \ln 2$? That would be an experiment on a scale where $T_{\text{equiv}}$ is far from the bath temperature.

# Chapter 10

## Landauer Bound: Phase Space, Not Semantics

> Physics counts regions. It does not read them.
>
> A271

### 10.1 Liouville and the Region

The Liouville theorem: The phase space volume of a closed system is invariant. When a region mapping $f : \{A, B\} \to Z$ reduces several initial regions to one target region, the "missing" volume must go somewhere. The bath takes it over. [B]

That is the physical core of the Landauer limit — without semantics, without the concept of information.

### 10.2 Three Region Operations

**Bijective mapping:** $W_{\text{after}} = W_{\text{before}}$. No costs. Transistor flips from A to B — a bijective operation. No volume loss. [B]

**Non-bijective mapping:** $W_{\text{after}} < W_{\text{before}}$. Minimum costs: $Q \geq k_B T \ln(W_{\text{before}}/W_{\text{after}})$. RESTORE TO ONE — the classical bit erasure. [B]

**Decoherence:** The superposition region $\{|\psi\rangle\}$ is reduced to a classical region. That is also a non-bijective region mapping. The costs fall to the bath — exactly Landauer. [B]

***Intuition:*** *Bijective is like rearranging furniture — same objects, different arrangement, nothing lost. Non-bijective is like pushing furniture together — a corner of the room remains free, and the room has not gained more space: The freed part must reappear somewhere else.*

## 10.3 The Reversal Test

Doc. 301 formulates the **reversal test** as an operational criterion: An operation is exactly then subject to Landauer costs when its reversal is *physically* not feasible — without involving a bath.

That is sharper than the usual formulation: Not the logical irreversibility (Landauer 1961) decides, but the physical non-bijectivity of the region mapping. [B Doc. 301]

## 10.4 Unit Cells and Spectral Response

When the phase space is divided into unit cells $h^f$, each cell gives exactly one orthogonal answer. The spectral structure of the Landauer bookkeeping is thus completely determined by $h^f$ and the ratio $W_{\text{before}}/W_{\text{after}}$ — no free parameters. [B Doc. 302]

# Chapter 11

## Information as Topologically Frozen Energy

> Information is topologically frozen kinetic energy.
>
> Doc. 255

### 11.1 Two Types of Energy in $\tilde{T} \cdot m = 1$

The fundamental relation of the FFGFT $\tilde{T} \cdot m = 1$ enforces a natural division of energy: static energy as topologically enclosed kinematics and free kinetic energy as motion through $T^4$.

**Static energy:** Energy bound to rest mass $E_0 = mc^2$ corresponds to topologically stable winding numbers on $T^4$. It is discrete, does not contribute to thermodynamic entropy, and remains conserved as long as the topology remains intact.

**Dynamic kinetic energy:** Free kinetic energy — for massless particles the only form of energy ($E = hf$), for massive particles the part beyond the rest mass. It is continuous, thermodynamic and entropy-producing.

### 11.2 Information as Frozen Kinematics

From this division follows without additional postulates:

**Kernresultat**

Information is topologically frozen kinetic energy. A winding configuration $(n_\theta, n_\varphi)$ on $T^4$ is a kinetic energy that is secured against dissipation by the topology of the torus. It cannot be lost without the topology breaking — and topological transitions are quantized in FFGFT.

Consequence: Landauer's principle describes the price for the *thawing* of frozen kinematics. The erasure of a bit is the enforced transfer of a topologically secured configuration into another — the phase space content released in the process goes to the bath.

## 11.3 Quantization of Information

The discreteness of the winding numbers enforces the quantization of information. There is no half bit — the winding number is an integer. That is not a postulate, but a direct consequence of the torus topology.

The winding quantum $\Delta w = 1$ is thus the fundamental information unit of the FFGFT — smaller is not possible, because the topology knows no non-integer winding numbers.

## 11.4 M. M. Vopson from an FFGFT Perspective

Vopson's equivalence **[V1]** $E_{\text{info}} = mc^2_{\text{Bit}}$ can be derived from FFGFT — but only for the special case of a specific scale. The energy of a winding quantum on scale $L$ is $\hbar c/L$, and the corresponding equivalent mass is $m = \hbar c/(Lc^2) = \hbar/(Lc)$. That is not a universal bit mass — it is scale-sensitive.

Vopson's **[V1]** result is the limiting case $L \to L_P$ (Planck scale): then $m_{\text{bit}} \to m_P$, the Planck mass. Matzke's derivation $m_{\text{bit}} = m_P \ln 2/(2\pi)$ is the Landauer special case at Planck temperature. Both are limiting cases of the FFGFT formula, not universal laws.

## 11.5 The Casimir Effect as Evidence

The Casimir effect shows that the vacuum itself carries information: The vacuum energy density at the characteristic $\xi$ length $L_\xi$ matches the Casimir energy density. That is no coincidence — it is the FFGFT prediction that $\xi$ determines both the geometric scale of the torus windings and the vacuum energy scale (Doc. 025).

# Chapter 12

## Thermodynamics and FFGFT

> Thermodynamics is its own bookkeeping level. Each level has its own currency.
>
> A283

### 12.1 Two Bookkeepings

FFGFT and thermodynamics are complementary levels of description of the same physical event. They speak different languages: [B]

| | FFGFT | Thermodynamics |
|---|---|---|
| Basic quantity | Winding number $w$, scale $L$ | Volume $W$ in $h^f$ units |
| Energy | $\hbar c/L$ per winding quantum | $k_B T \ln(W_1/W_2)$ |
| Irreversibility | Geometric (torus) | Statistical (Liouville) |
| Universality | Scale-sensitive | Content-neutral |

Contact: No direct translation of winding numbers into entropy. Thermodynamics counts regions, FFGFT counts windings. The regions are related to the geometry via $h^f$ — that is the only bridge. [B]

***Intuition:*** *Two accountants audit the same company: one in euros, one in dollars. They arrive at the same result, but their books look different. Thermodynamics and FFGFT audit the same physics — with different currencies. The conversion point is $h^f$.*

### 12.2 Entropy in the FFGFT Picture

Entropy is not a fundamental quantity in FFGFT — it is a statement about how many geometrically different configurations a system can have on a given scale:

$$S = k_B \ln W = k_B \ln\left(\frac{L^f}{h^f}\right)$$

For $f = 2$ (torus section) and $L = L_N$: direct connection to the $\xi$ hierarchy (Doc. A015, A283). [K]

## 12.3 Black Holes as a Limiting Case

A black hole in the FFGFT has maximum entropy for a given volume — that is geometric, not thermodynamically primary. The Bekenstein-Hawking entropy $S_{\mathrm{BH}} = A/(4\ell_{\mathrm{P}}^2)$ is, in this picture, the maximum winding number on the Schwarzschild scale. [B]

Hawking radiation as a non-bijective region mapping: the region inside is reduced to a target region outside. In the process, Landauer costs arise. The complete derivation without a universal bit value is in Doc. 325 (§ 15). [B]

# Part IV

# Comparison and Classification

# Chapter 13

# Hyperbit vs. FFGFT: A Structural Comparison

> Both calculations work out. They only look elegant at different points.
>
> Doc. 326

## 13.1 Starting Points in Comparison

**Matzke/Hyperbit**

[SETZUNG] $e_i^2 = +1$ (one axiom).
Borrowed bridges: Bekenstein, Compton = Schwarzschild, Landauer at $T_P$.
Result: Black hole physics from combinatorics, $m_{\text{bit}} = m_P \ln 2/(2\pi)$.

**FFGFT**

[SETZUNG] $\xi_{\text{par}} = 4/30000$, torus $T^4$, winding quantization.
No borrowed bridges — but higher entry price.
Result: Bit energy is scale-sensitive, $E_{\text{bit}} = \hbar c/L$. [K]

***Intuition:*** *Two hikers want to reach the same summit. One takes the short, steep route — he needs three ropes from others (Bekenstein, Compton, Landauer). The other builds the entire path himself, pays the price upfront, and needs no help afterwards. Both arrive. But only one has truly managed without foreign material.*

## 13.2 Convergence: Where Both Agree

Both frameworks agree: [B]

- Geometry arises from distinguishability.
- Fermion generations depend on dimension-3 structures.

- Black holes mark an information threshold.
- The Planck regime is the natural limiting case.

## 13.3 Divergence: The Landauer Point

The essential difference: Matzke sets a universal bit mass; FFGFT shows that $E_{\text{bit}}$ depends on the scale. [K]

That is not only quantitative — it is conceptual: If $m_{\text{bit}}$ is universal, then all bits have the same physical status, regardless of whether they are encoded in a DRAM, a spin or a photon. If $E_{\text{bit}} = \hbar c/L$, then *there is no system-independent bit mass* — the number always depends on a physical carrier with a physical scale.

## 13.4 Cl(6) and GF(27)

Doc. 326 and 341 compare the algebraic structures: Cl$(6)$ with $\mathbb{Z}_3 \times \mathbb{Z}_3$ symmetry corresponds in the FFGFT to the Galois structure GF$(27)$ = GF$(3^3)$. Three colors, three generations — this three is geometrically grounded in both the hyperbit picture and the FFGFT picture. [B]

Open Question [S]: Is Cl$(6)$ a special case of the FFGFT Galois structure on $T^4$, or are both independent paths to the same physics?

# Chapter 14

## Time in State Space or Outside — Origin of Mass Modes

> Is time a direction in the state space, or a parameter beside it? This question sounds technical, but it is ontological.
>
> Doc. 307

### 14.1 The Fork

At the beginning of every quantum mechanical model building stands a choice that usually remains unspoken: **Is time a direction *in* the state space, or a parameter *beside* it?**

The difference lies not in the mathematics used — both answers work with self-adjoint operators, spectra and projectors — but in *where* time sits. And precisely this placement decides whether the physical quantities *fall* out of the structure or must be *attached* to it.

### 14.2 The Unrolled View: Time as an External Parameter

The familiar formulation of quantum mechanics locates time outside the state space. The Hilbert space is the space of states *at one time*. Time is the parameter $t$ along which the state evolves: $e^{-i\hat{H}t}$. The observables are operators *on* $\mathscr{H}$. Time is not one — Pauli's objection shows that for a Hamiltonian bounded below, no self-adjoint, canonically conjugate time operator exists.

Consequence: Since time stands outside, the state space alone contains no dynamic scales. Masses, couplings, lifetimes must be introduced from outside via *additional* structures. In the Standard Model, this happens through Yukawa terms with free parameters. Geometry provides the framework; the numbers come from measurement.

## 14.3 The Rolled-Up View: Time as a Compact Dimension

FFGFT pulls time *into* the state space as a closed, periodic dimension. The Hilbert space is $\mathscr{H} = L^2(T^4) \otimes \mathbb{C}^3$, where one of the torus dimensions carries time.

In this view, the masses become *eigenvalues of a mode structure on the compact torus*:

$$\tilde{T} \cdot m = 1 \quad \text{on } L^2(T^4) \otimes \mathbb{C}^3$$

The mass ratios are thus exact and derived from a single dimensionless parameter $\xi = 4/30000$. The absolute scale carries its SI correction via fixed bridge constants, but is not a free parameter.

## 14.4 Why the Unrolled View Does Not Explain Mass

A parametric mapping with free parameters can hit any finite list of values. It possesses no predictive content as long as it is not *fixed in advance and universally*. To deliver numbers, it must be coupled to external values — ultimately to the Standard Model itself. That is the difficult path: The numbers come from outside, not from the structure.

The point is not that this procedure would be wrong — it is honest, provided it declares its own limit. The point is that the difficulty is not a weakness of the model, but the immediate consequence of the unrolled placement of time.

## 14.5 Connection to the Hyperbit Discussion

Matzke's hyperbit algebra sits in the unrolled view — time stands outside, the Clifford algebra Cl$(6)$ provides the algebraic structure (three colors, one generation). What hyperbit lacks is precise to name: **no SI anchor**, no absolute mass scale from the structure itself. [B Doc. 307]

This is a different situation from the Standard Model. The Standard Model has 19 free parameters — it depends entirely on measurement. Hyperbit has the structure algebraically, but must introduce the mass via a bridge (Landauer at $T_P$). The bridge is a stipulation, not a free parameter in the Standard Model sense.

***Intuition:*** *Imagine three levels. At the bottom: the Standard Model — 19 measurements, 19 hooks from outside. In the middle: Hyperbit — the algebraic structure stands, but one hook is still missing: the absolute mass scale. At the top: FFGFT — the ratios fall out internally from $\xi$ and GF($3^n$); the absolute scale needs a single fixed bridge constant (v = 246 GeV), which is not a free measurement.* [K]

**The algebraic structure** Cl$(6)$ $\leftrightarrow$ GF$(27)$ is translatable in FFGFT (Doc. 326/341). This translation shows: Hyperbit and FFGFT describe the same threefold structure on different algebraic paths. [B] What Hyperbit would gain in this translation: the absolute mass scale from the mode structure, without the Landauer bridge.

**Doc. 307**

**Standard Model** (unrolled): Time outside, 19 free parameters, all masses from outside.

**Hyperbit** (unrolled, structured): Time outside, algebraic structure internal, but SI anchor missing — one bridge needed.

**FFGFT** (rolled-up): Time inside ($\tilde{T} \cdot m = 1$), mass ratios internal from $\xi$ and GF($3^n$), one fixed bridge constant for the absolute scale.

# Chapter 15

## Hawking and the Information Paradox in the FFGFT

> The paradox does not resolve by adding new physics, but by clarifying the premises.
>
> Doc. 250

### 15.1 The Paradox and Its Three Premises

Hawking showed in 1974 that black holes emit thermal radiation. This radiation carries no information about the state of the infalling matter — it is purely thermal. If the black hole completely evaporates, the original quantum information would be destroyed. That contradicts the unitarity of quantum mechanics.

The paradox presupposes three premises, which do not all hold in FFGFT:

1. Information is the complete description of the quantum state of all infalling particles — an *epistemic* category.
2. Time is continuous and one-dimensionally directed.
3. The classical singularity is physically real.

FFGFT rejects all three — not by assertion, but through the geometric structure of the framework.

### 15.2 Fundamental Information vs. Epistemic Information

**Kernresultat**

The fundamental information of a physical system in FFGFT is the **topological charge** of its constituents: the winding numbers $(n_\theta, n_\varphi, n_3, n_4)$ on $T^4$. These define *which kind* of particles exist — not where they are, not in which exact quantum state they are.

Epistemic information — what an observer can reconstruct — can be lost through thermalization. That is thermodynamic irreversibility, not ontological information loss.

The Hawking paradox is an *epistemic* problem: It asks whether an observer outside the horizon can completely reconstruct the original quantum state. FFGFT answers a different question: Does the fundamental topological structure continue to exist?

## 15.3 The $T^4$ Compactification Undermines Premise 2

In FFGFT, time is a compact dimension of the $T^4$ torus, not a continuous external parameter. The compactification means: There is no absolute 'forever inaccessible' — the time coordinate is periodic. What classically appears as irreversible information loss behind the horizon is, in FFGFT, a geometric projection, not ontological annihilation.

## 15.4 The Singularity is Eliminated: Premise 3

In FFGFT there is no singularity in the classical sense. The minimal core radius $L_0 = \xi \cdot \ell_\text{P}$ prevents complete collapse. The winding numbers remain conserved — the topological charge is indestructible. A black hole in FFGFT is not an information destroyer, but a state of maximum winding density on the Schwarzschild scale.

## 15.5 Hawking Temperature Without a Universal Bit Value — Doc. 325

Doc. 325 shows: The complete Hawking physics — temperature, spectrum, information conservation — follows from $\tilde{T} \cdot m = 1$, the $\mathbb{Z}_3$ structure and the system-dependent bit energy $E_\text{bit} = \hbar c / L$, *without* a universal bit value.

That is the decisive difference from Matzke: Matzke sets $m_{\text{bit}} = m_P \ln 2/(2\pi)$ as a universal constant and derives the Schwarzschild radius from it. FFGFT arrives at the same numerical results — but without this stipulation. The stability threshold $n_{\text{thresh}} = 6{,}41$ in Matzke depends on the bit value parametrization, not on the temperature physics.

**Kernresultat**

Matzke and FFGFT converge on **fermion generation = fundamental algebraic unit**. They diverge on **bit values**: universal (Matzke, Landauer at $T_P$) vs. system-dependent (FFGFT, $E_{\text{bit}} = \hbar c/L$). This difference concerns the stability threshold and the mass of a bit. The Hawking temperature formula itself is unaffected: Both routes give the same number, but only the FFGFT route manages without a universal bit value.

# Chapter 16

## Information, Measurement and Projection

> Measuring is a physical process, not an epistemological one.
>
> A284

### 16.1 What is a Measurement?

In the FFGFT, a measurement is a **projection**: The torsion field is projected onto a state that is orthogonal-stable to the measuring apparatus. That is a region reduction in phase space — and thus a Landauer process. [B]

***Intuition:*** *Imagine a shadow on the wall. The three-dimensional object is projected onto a two-dimensional surface. Information is lost — height, depth, angle. A measurement is similar: The quantum mechanical state is projected onto a classical measured value. The remaining possibilities disappear into the bath.*

### 16.2 Projection as a Non-Bijective Mapping

Quantum measurement is non-bijective: From the states $\{|\psi_1\rangle, |\psi_2\rangle, \ldots\}$ one state $|k\rangle$ is selected. The volume of the remaining states must go somewhere — it goes to the bath (measuring apparatus and environment). That is Landauer. [B]

That is the physical reason why quantum measurements are fundamentally irreversible: not because of epistemology, but because of Liouville.

### 16.3 Magnitude and Phase

Doc. 290 poses the question: What is determined in a measurement? Magnitude or phase?

The answer depends on the measuring apparatus, not on the measured system. The *information* (in the abstract sense) is magnitude and phase together. The measurement selects one component — that is a descriptive decision at Level 3 — and costs the corresponding region volume to the bath.

Open Question [S] (Doc. 290): Is the phase information loss in measurement thermodynamically equivalent to the magnitude information loss? The region volumes are different — but the temperature is the same. This possibly gives a measurable difference in dissipation.

# Chapter 17

## The Computing Marble

> Visibility of the structure and binding force of the number run counter to each other. The more primitive the carrier, the clearer the bookkeeping and the more irrelevant the limit.
>
> A273

### 17.1 The Supply as a Visible Bath

A271 works with the world ocean — the bath is developed, but invisible. When computing with tokens — abacus beads, shells, coins — there is a **supply** next to the working area: tokens come from it, and into it they flow back. The supply is a bath that one can point to.

The structure is the same as in A271: [B]

- **Region:** A token left ≠ token right — two distinguishable macro-regions.
- **Region reduction:** Resetting the board maps all configurations onto one — non-bijective, subject to Landauer costs.
- **Bath:** The supply takes over the volume.

***Intuition:*** *An abacus has beads and a frame. Resetting the abacus means: all beads to the left — no matter what was there before. That is a non-bijective region mapping. The costs are real, but tiny: about $5 \cdot 10^{15}$ times the Landauer limit. Visible — but thermodynamically irrelevant.*

### 17.2 The Limiting Transition: To the Brownian Sphere

If the token shrinks until its barrier approaches $k_B T$, it becomes the **Brownian computing sphere** — that is Bérut's colloid experiment (2012), which experimentally confirmed Landauer's limit. [Q]

The limiting transition is continuous: Abacus → nano-magnet → colloid → qubit. There is no qualitative jump between macroscopic computing and quantum physical computing — only a quantitative transition in the barrier height relative to $k_B T$. [B]

***Intuition:*** *It is like the transition from a huge boulder (classical bit) to a grain of sand (p-bit) to a single molecule (qubit). The same physical process, only on different scales. The bookkeeping is the same in all cases — Liouville, Landauer, bath.*

# Part V

# The Bigger Picture

# Chapter 18

## $\xi$ from Galois Algebra — No Free Constant

> $\xi$ is not a free constant — it follows necessarily from the Galois group orders and the winding numbers of the $T^4$ torus.
>
> Doc. 338

### 18.1 The Core Identity

The square of the lepton mass ratio yields an exact integer identity that both sides — FFGFT and Galois algebra — simultaneously satisfy ():

$$\boxed{\underbrace{\frac{(r_\mu/r_e)^2}{\xi}}_{\text{FFGFT}} = \underbrace{|\mathrm{GF}(9)^*|^2 \cdot 5^2 \cdot |\mathrm{GF}(27)|}_{\text{Galois}} = 8^2 \cdot 25 \cdot 27 = 43200}$$

The three Galois factors have concrete algebraic significance:

- $8 = |\mathrm{GF}(9)^*| = 3^2 - 1$: multiplicative group of the 2nd generation
- $27 = |\mathrm{GF}(27)| = 3^3$: field of the 3rd generation (all elements)
- $5 =$ smallest prime divisor of $|\mathrm{GF}(81)^*| = 80$: first new prime in $\mathrm{GF}(3^4)^*$

### 18.2 Derivation of $\xi$ from the Identity

Since both sides are the same number, $\xi$ follows algebraically:

$$\xi = \frac{(r_\mu/r_e)^2}{43200} = \frac{(12/5)^2}{43200} = \frac{144/25}{43200} = \frac{1}{7500} = \frac{4}{30000}$$

The winding numbers ($r_e = 4/3,\ r_\mu = 16/5$) in turn follow from the Galois group structures of $\mathrm{GF}(3^n)$:

| Quantity | Value | Galois Origin | Remark |
|---|---|---|---|
| $n_{\theta,1}$ | 3 | char(GF$(3^n)$) | axiomatic |
| $n_{\phi,1}$ | 2 | \|GF$(3)^*$\| | mult. group of the prime field |
| $n_{\theta,2}$ | 5 | recursion $3+2$ | smallest prime divisor \|GF$(81)^*$\| $= 80$ |
| $n_{\phi,2}$ | 4 | $\varphi(5)$ | Euler $\varphi$ of the prime divisor |
| $n_{\theta,3}$ | 9 | recursion $5+4$ | simultaneously char$^2$ |
| $n_{\phi,3}$ | 5 | smallest prime divisor | |

Recursion law: $n_{\theta,g} = n_{\theta,g-1} + n_{\phi,g-1}$.

## 18.3 Connection to $\alpha$

The geometric correction factor $K_{\text{frak}}$ cancels exactly in mass ratios. It connects $\xi$ with the fine structure constant (Doc. 070):

$$\alpha = \frac{\xi \cdot E_0^2}{K_{\text{frak}}}, \quad E_0 = \sqrt{m_e m_\mu} \approx 7{,}35 \text{ MeV}$$

$$\Rightarrow \quad \alpha^{-1} = 137{,}06 \quad \text{(Dev. 0,02 \%)}$$

Thus $\xi$ is anchored twice:

1. **Algebraically** from the Galois group orders GF$(3^n)$ and the torus winding numbers — purely structural, no free parameter.
2. **Empirically** through $\alpha$: A single measured value ($\alpha \approx 1/137{,}036$) anchors $\xi$ in physical reality.

## 18.4 Comparison: FFGFT vs. Galois vs. Experiment

| Method | $(m_\mu/m_e)^2$ | $m_\mu/m_e$ | Dev. exp. |
|---|---|---|---|
| Experiment (PDG) | 42 753 | 206,768 | — |
| FFGFT bare | 43 200 | 207,846 | $+0{,}52$ % |
| Galois $8^2 \cdot 25 \cdot 27$ | 43 200 | 207,846 | $+0{,}52$ % |

FFGFT and Galois give exactly the same bare value. The 0,52 % deviation from experiment is the same fractal-recursive higher-order correction in both frameworks — it is already fully contained in the PDG measured values.

## 18.5 Significance for the Hyperbit Discussion

Matzke's hyperbit begins with $e_i^2 = +1$ — a structural stipulation whose justification remains open. The FFGFT counterpart is not $\xi$ (which is algebraically derived), but the choice of the field GF$(3^n)$ as the fundamental structure. This choice is motivated by the three-generation structure of fermions, but is not yet a complete derivation.

The decisive difference: In the FFGFT, the step from the algebra (GF$(3^n)$, winding numbers) to the numerical value $\xi$ is closed. In Matzke's framework, the step from $e_i^2 = +1$ to the bit mass ($m_{\text{bit}} = m_P \ln 2/(2\pi)$) remains a borrowed bridge via Landauer at Planck temperature — no algebraic derivation.

# Chapter 19

## Everything Is Information — But Which?

> If one sets all units to 1, ratios remain. Pure numbers. Structures. Information.
>
> Matrix Book, Ch. 16

### 19.1 The Fourth Reading

The T0 theory knows three equally valid readings: Everything is energy, everything is mass, everything is time. There is a fourth:

**Everything is information.**

What remains invariant when one sets all units to 1 ($c = \hbar = \alpha = G = 1$)? **Ratios** remain. Pure numbers. Structures. The mass ratio $m_e/m_\mu \approx 1/206{,}8$ is not energy — it is a relation, a piece of pure information about the geometry of the torus. [B]

### 19.2 Information Without a Carrier?

Here lies the decisive tension with A271–A273:

In the Landauer analysis, information is *always* bound to a carrier. A bit without a carrier is an abstract category, not a physical quantity — and has no energy. [B]

In the cosmological reading, the ratios $\xi, m_e/m_\mu, G/\hbar c$ are *self-supporting* information — they are not encoded in an external medium. The universe is its own carrier.

That is not a contradiction, but a limiting-case situation: When the medium *itself* consists of information, there is no longer an independent carrier.

***Intuition:*** *A book contains information on paper — the carrier is the paper. But what if the paper itself consists of letters? Then the carrier and the content are the same. That is the situation in the universe, if one understands it as a geometric structure: The geometry is the carrier medium and the information simultaneously.*

## 19.3 $\xi$ from the Galois Algebra

In the FFGFT, $\xi = 4/30000$ is not an open question. It follows necessarily from the Galois group orders of the field GF$(3^n)$ and the winding numbers of the $T^4$ torus (Doc. 338):

$$\xi = \frac{(r_\mu/r_e)^2}{|\mathrm{GF}(9)^*|^2 \cdot 5^2 \cdot |\mathrm{GF}(27)|} = \frac{(12/5)^2}{8^2 \cdot 25 \cdot 27} = \frac{4}{30000}$$

The three Galois factors have concrete algebraic significance: [K Doc. 338]

$8 = |\mathrm{GF}(9)^*| = 3^2 - 1$ (multiplicative group of the 2nd generation), $27 = |\mathrm{GF}(27)| = 3^3$ (field of the 3rd generation), $5 =$ smallest prime divisor of $|\mathrm{GF}(81)^*| = 80$.

Additionally, $\xi$ is empirically anchored (Matrix Book, Ch. 2): $\xi = \alpha/(E_0/\mathrm{MeV})^2$.

The answer to the "why" of $\xi$ is thus algebraic, not speculative. In Matzke's framework, the analogous question — Why $e_i^2 = +1$? — remains genuinely open.

## 19.4 Where Hyperbit and T0 Converge

Matzke's hyperbit: 'There are distinguishable things' → Geometry.

FFGFT: 'There are windings on $T^4$' → Geometry, physics, masses.

Both start with pure distinguishability — pure information — and end at physics. That is the deepest common point of both approaches. [B]

# Chapter 20

## Open Questions and Research Directions

A theory that has no open questions has ceased to research.

### 20.1 Open Questions in the Hyperbit–FFGFT Connection

**Cl(6) and GF(27).** Is the Clifford algebra Cl$(6)$ a special case of the FFGFT Galois structure on $T^4$, or are both independent paths to the same physics? If there is a mapping: What are the translation rules?

**Universal bit mass.** Matzke's $m_{\text{bit}} = m_{\text{P}} \ln 2/(2\pi)$ is a universal constant — in the FFGFT it is scale-sensitive. Is there a range in which both predictions are experimentally distinguishable?

**p-bit prediction.** FFGFT says: natural p-bit materials at $L_8 \approx 22$ nm. Are there measurements that show this scale as particularly stable or unstable?

### 20.2 Open Questions in the Landauer Interpretation

**Multiple-realization invariance (Doc. A272).** Are there physical systems in which the invariance breaks — systems in which the energy varies with the information content through constructive coupling? That would touch the proof of the carrier theorem, but not its premise.

**Phase dissipation.** Is the information loss in phase during a quantum measurement thermodynamically equivalent to the magnitude information loss? (Doc. 290)

**Continuous entropy flow.** DRAM refresh costs: How exactly does the Landauer limit match measured entropy flows in modern processors? Are there systematic deviations due to quantum corrections?

## 20.3 Open Question: Matzke's Axiom

**Why $e_i^2 = +1$?** In Matzke's hyperbit framework, it remains open why exactly this sign forms the basis of physics. In the FFGFT, the analogous question is answered: $\xi = 4/30000$ follows algebraically from the Galois group orders (Doc. 338). Matzke's open question has no direct counterpart in the FFGFT.

## 20.4 Experimental Approaches

Three experimental lines are particularly promising:

1. **Bérut-type experiments** at the p-bit scale: Can one directly measure Landauer's limit at $L_8 \approx 22$ nm and room temperature?
2. **Qubit decoherence calorimetry**: Does the heat generation during decoherence match the Landauer formula?
3. **Galois spectroscopy**: Does GF$(27)$ structure in particle spectra (Doc. 356–357) show measurable signatures?

## 20.5 Status Balance

| Statement | Status |
|---|---|
| Bit ≠ physical region (Shannon vs. Boltzmann) | [B] |
| $S = k_B \ln W$ as starting point | [SETZUNG] |
| Region reduction ⇒ Landauer limit | [B] |
| Liouville theorem | [B] |
| Content neutrality of thermodynamics | [B] |
| $h^f$ as fundamental discretization | [B] |
| Hardware ≠ computational operation | [B] |
| Energy depends on the carrier, not the information | [B] |
| Permanent memory as counter-experiment to Vopson | [B] |
| $E_{\text{bit}} = \hbar c/L$ (scale-sensitive) | [B] |
| $\xi = 4/30000$ from Galois group orders | [K] |
| Matzke's $m_{\text{bit}} = m_{\text{P}} \ln 2/(2\pi)$ (Landauer at $T_P$) | [SETZUNG] |
| Topological information conservation in FFGFT | [K] |

| Statement | Status |
|---|---|
| $n_{\text{thresh}} = 6{,}41$ system-dependent (not universal) | [K] |
| $T^4$ compactification solves Hawking paradox | [B] |
| Matzke's question $e_i^2 = +1$: open | [H] |
| Multiple-realization invariance: open | [H] |
| Bérut-type experiment at p-bit scale | [S] |

# Appendices

## Layer Markers and Conventions

### Layer Markings

Every substantive statement in this book carries a layer marking. This convention originates from the FFGFT corpus (Doc. A010):

[SETZUNG] Declared starting point. Axiom, stipulation, not provable within the system — but explicitly named as such.

[B] Algebraically/mathematically derived from declared foundations. Formal derivation step.

[Q] Source. External primary source or measured value. Can be verified with the corpus.

[S] Plausibility sketch. Argument that convinces, but is not formally proven.

[H] Open research question. Explicitly not answered.

[K] Core result. Summary that is stand-alone readable even without the proof.

### Mathematical Conventions

- $\xi_{\text{par}} = 4/30000$ — FFGFT fundamental parameter
- $\ell_{\text{P}} = \sqrt{\hbar G/c^3}$ — Planck length
- $m_{\text{P}} = \sqrt{\hbar c/G}$ — Planck mass
- $E_P = m_{\text{P}} c^2$ — Planck energy
- $T_P = E_P/k_B$ — Planck temperature
- $E_{\text{bit}} = \hbar c/L$ — Bit energy on scale $L$ (FFGFT)
- $m_{\text{bit}} = m_{\text{P}} \ln 2/(2\pi)$ — Bit mass (Matzke)
- $W$ — Phase space volume in units of $h^f$
- $f$ — Dimension of the phase space

### Test Scripts

The numerical statements of the primary documents are backed by Python test scripts with assertions:

- `a271_landauer.py` — Checks 1–10 (Doc. A271)
- `a273_rechenkugel.py` — Checks 1–14 (Doc. A273)

All scripts are available in the GitHub repository under `2/python/`.

## Source Documents

All documents are freely accessible on GitHub:
https://github.com/jpascher/T0-Time-Mass-Duality/

### A-Series (Information Physics, Landauer, Thermodynamics)

| Document | Topic |
|---|---|
| A180 | Information (Foundations) |
| A271 | Landauer and Phase-Space Bookkeeping |
| A272 | Carrier and Information |
| A273 | The Computing Sphere |
| A280 | Landauer Phase-Space Bookkeeping (Extended) |
| A281 | Carrier and Information (Extended) |
| A282 | The Computing Sphere (Extended) |
| A283 | Thermodynamics and FFGFT |
| A284 | Measurement and Projection |

### Main Corpus (Doc. 100–360, Thematic Selection)

| Document | Topic |
|---|---|
| 174 | Spin and Qubit States |
| 184 | The p-Bit |
| 243 | FFGFT Information Formalism |
| 247 | Information, State, Process |
| 255 | Information Kinetics |
| 257 | Information Unit and Scale ($E_{\text{bit}} = \hbar c/L$) |
| 290 | Information Question: Magnitude and Phase |
| 301 | Information Reversal Test |
| 302 | Unit Cells and Bit Capacity |
| 324 | G(6,$\mathbb{Z}_3$C)–FFGFT Comparison |
| 325 | Hawking and FFGFT |
| 326 | Matzke–FFGFT Comparison |

## References

### Matzke Sources

**[M1]** D. Matzke, *Black Holes from Hyperbits*, IPI Annual Conference, August 2026.
Main source for the hyperbit framework; derivation of Schwarzschild radius and Bekenstein entropy from $e_i^2 = +1$.

**[M2]** D. Matzke, *G(6, $\mathbb{Z}_3$C) Hyperspin Framework*, IPI Working Paper, 2026.
Algebraic foundation; Cl$(6)$ and fermion generations. Compared with FFGFT in Doc. 324 (PDF).

### FFGFT Corpus (J. Pascher)

**[P1]** J. Pascher, *A271 — Landauer and Phase-Space Bookkeeping*, FFGFT Corpus, July 2026.
Direct Link PDF · Core source for Ch. 3 and 10.

**[P2]** J. Pascher, *A272 — Carrier and Information*, FFGFT Corpus, July 2026.
Direct Link PDF · Textual criticism of the Landauer reception; Ch. 5.

**[P3]** J. Pascher, *A273 — The Computing Sphere*, FFGFT Corpus, July 2026.
Direct Link PDF · Abacus, colloid, limiting transition; Ch. 17.

**[P4]** J. Pascher, *A280/A281 — Landauer Phase-Space Bookkeeping (Extended) / Carrier and Information (Extended)*, FFGFT Corpus, 2026.
A280 PDF · A281 PDF.

**[P5]** J. Pascher, *A283 — Thermodynamics and FFGFT*, FFGFT Corpus, 2026.
Direct Link PDF · Ch. 12.

**[P6]** J. Pascher, *A284 — Measurement and Projection*, FFGFT Corpus, 2026.
Direct Link PDF · Ch. 16.

**[P7]** J. Pascher, *Doc. 257 — Information Unit on $T^4$*, FFGFT Corpus.
Direct Link PDF · $E_{\text{bit}} = \hbar c/L$, winding quantum $\Delta w = 1$; Ch. 9.

**[P8]** J. Pascher, *Doc. 184 — The p-Bit*, FFGFT Corpus.
Direct Link PDF · $L_8 \approx 22$ nm, Stoner-Wohlfarth; Ch. 5.

**[P9]** J. Pascher, *Doc. 326 — Matzke–FFGFT Comparison*, FFGFT Corpus, August 2026.
Direct Link PDF · Basis for Ch. 10.

**[P10]** J. Pascher, *Doc. 324 — G(6, $\mathbb{Z}_3$C)–FFGFT Comparison*, FFGFT Corpus, 2026.
Direct Link PDF.

**[P11]** J. Pascher, *Doc. 325 — Hawking and FFGFT*, FFGFT Corpus, 2026.
Direct Link PDF.

**[P12]** J. Pascher, *Doc. 302 — Unit Cells and Bit Capacity*, FFGFT Corpus, 2026.
Direct Link PDF.

**[P13]** J. Pascher, *Doc. 301 — Information Reversal Test*, FFGFT Corpus, 2026.
Direct Link PDF.

**[P14]** J. Pascher, *Doc. 174 — Spin and Qubit States*, FFGFT Corpus.
Direct Link PDF.

## Mathematical Foundations (External Sources)

**[T1]** A. Hatcher, *Algebraic Topology*, Cambridge University Press, 2002.
Freely available: pi.math.cornell.edu/~hatcher/AT/AT.pdf
Ch. 1.2: Fundamental group of the circle $\pi_1(S^1) \cong \mathbb{Z}$; Ch. 1.2 Example 1.22: Product formula $\pi_1(X \times Y) \cong \pi_1(X) \times \pi_1(Y)$, from which $\pi_1(T^n) \cong \mathbb{Z}^n$ follows.

**[T2]** G. E. Bredon, *Topology and Geometry*, Springer GTM 139, 1993.
Ch. III.§5: Homotopy groups of product spaces.

**[T3]** J. R. Munkres, *Topology*, 2nd ed., Prentice Hall, 2000.
Ch. 9: Fundamental group; Ch. 11: Seifert-van Kampen theorem.

**[V1]** M. M. Vopson, *The mass-energy-information equivalence principle*, AIP Advances **9**(9), 095206 (2019).
Original formulation of the mass-information equivalence; basis for all Vopson discussions in Ch. 3, 4, 9.

**[V2]** M. M. Vopson, *The information catastrophe*, AIP Advances **10**(8), 085014 (2020).
Estimation of the bit capacity of the universe; $\sim 10^{80}$ to $10^{90}$ bits.

**[V3]** M. M. Vopson, *Second law of infodynamics and its implications for the simulated universe hypothesis*, AIP Advances **13**, 105308 (2023).
2nd law of infodynamics; symmetry minimization as an information principle (Hund's rules).

**[V4]** M. M. Vopson, *Experimental protocol to test the second law of infodynamics using a scanning tunnelling microscope*, AIP Advances **14**, 035205 (2024).
Experimental proposal for detecting infodynamics; dark matter as stored information.

**[B1]** R. Landauer, *Irreversibility and Heat Generation in the Computing Process*, IBM J. Res. Dev. **5**, 183 (1961).
Primary source; "logically irreversible" is the precise formulation.

**[B2]** Liouville theorem: e.g. Landau/Lifshitz, *Mechanics*, §46.

**[B3]** C. H. Bennett, *The Thermodynamics of Computation*, Int. J. Theor. Phys. **21**, 905 (1982); idem, *Notes on Landauer's Principle*, Stud. Hist. Phil. Mod. Phys. **34**, 501 (2003).

**[B4]** A. Bérut et al., *Experimental verification of Landauer's principle*, Nature **483**, 187 (2012).
Colloid experiment; directly confirms the Landauer limit.

**[B5]** Y. Jun, M. Gavrilov, J. Bechhoefer, *High-Precision Test of Landauer's Principle in a Feedback Trap*, Phys. Rev. Lett. **113**, 190601 (2014).

**[B6]** C. E. Shannon, *A Mathematical Theory of Communication*, Bell System Technical Journal **27**, 379 (1948).
Information-theoretic bit ($\log_2$), not thermodynamic.

**[B7]** E. T. Jaynes, *Information Theory and Statistical Mechanics*, Phys. Rev. **106**, 620 (1957).
Shannon-Boltzmann connection; applies exactly only for uniform distribution.

**[B8]** C. Furey, *Standard Model Physics from an Algebra?*, Diss., Waterloo, 2015.
Cl$(6)$ and fermion generations; foundation of **[M1/M2]**.

**[B9]** J. Bekenstein, *Black holes and entropy*, Phys. Rev. D **7**, 2333 (1973).
Entropy-area relation; one of the three bridges in **[M1]**.

**[B10]** T. Sagawa, M. Ueda, *Second Law of Thermodynamics with Discrete Quantum Feedback Control*, Phys. Rev. Lett. **100**, 080403 (2008).

**[B11]** E. Fredkin, T. Toffoli, *Conservative Logic*, Int. J. Theor. Phys. **21**, 219 (1982).
Reversible gates; Landauer-cost-free computing is in principle possible.

**[B12]** W. K. Wootters, W. H. Zurek, *A single quantum cannot be cloned*, Nature **299**, 802 (1982). No-cloning theorem.

**[B13]** P. W. Shor, Phys. Rev. A **52**, R2493 (1995); A. M. Steane, Phys. Rev. Lett. **77**, 793 (1996).
Quantum error correction.

## Use of AI Tools

This book was created using AI-supported tools. The questions, the stipulations and all substantive decisions come from the author; the tools formulate, calculate and create test scripts. Responsibility for content and errors lies entirely with the author.

**Specifically:** Structure, organization, derivation steps, layer markings ([SETZUNG], [B], [Q], [S], [H]) and all substantive decisions — what is asserted, what is considered open, what is rejected — come from the primary documents of the FFGFT corpus and from the author. AI tools have generated coherent prose and LaTeX code from them as well as in individual cases suggested formulations that the author has reviewed and adopted or rejected.

**Reference:** The detailed wording of this convention is in Doc. A010 (*Purpose and Structure of the A-Series*), A010 (PDF).